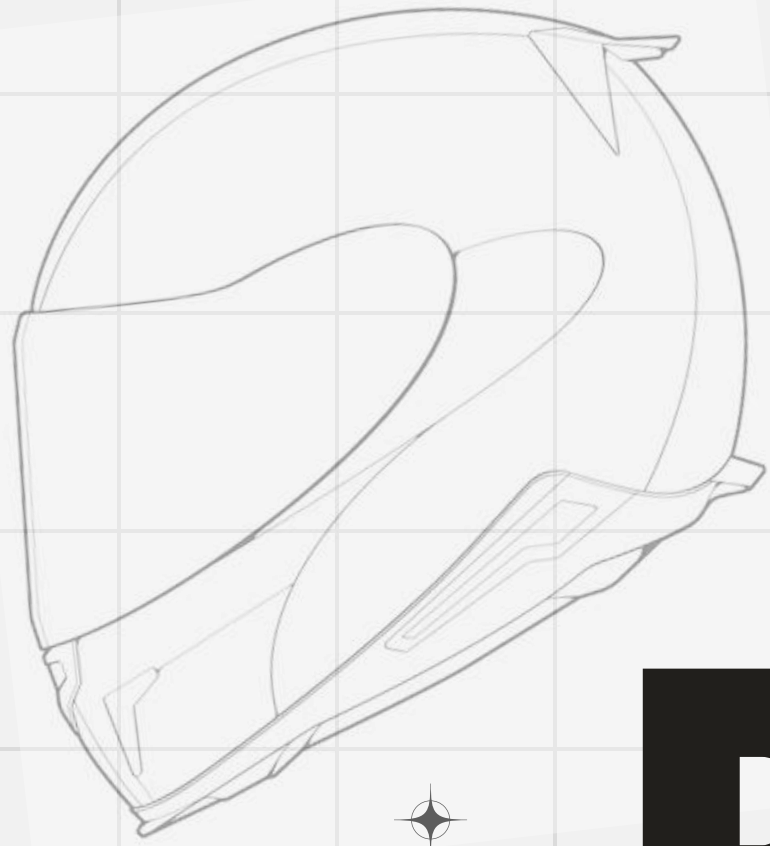
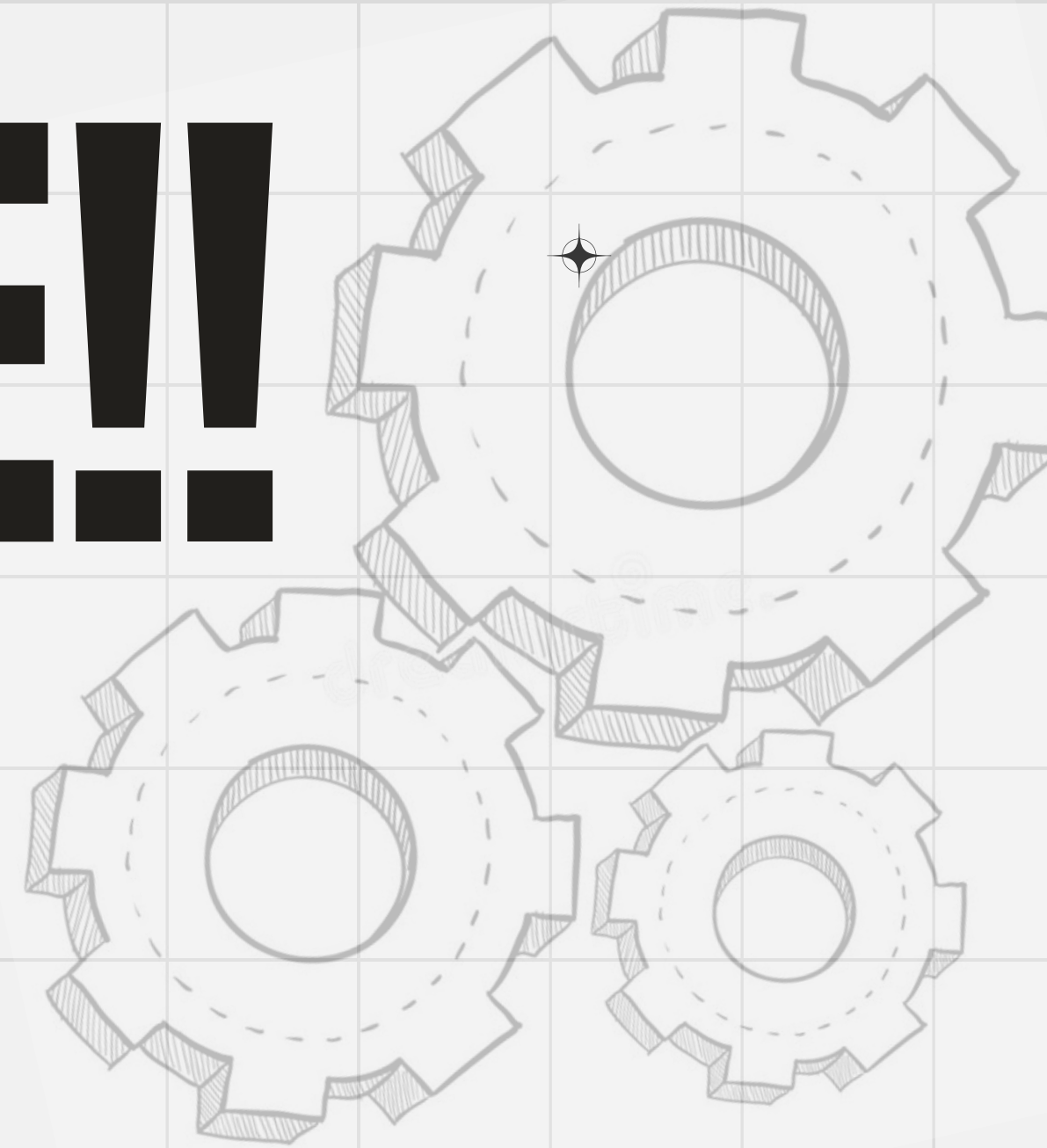
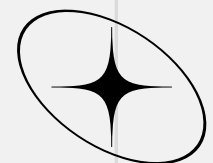




RIDE SAFE!!

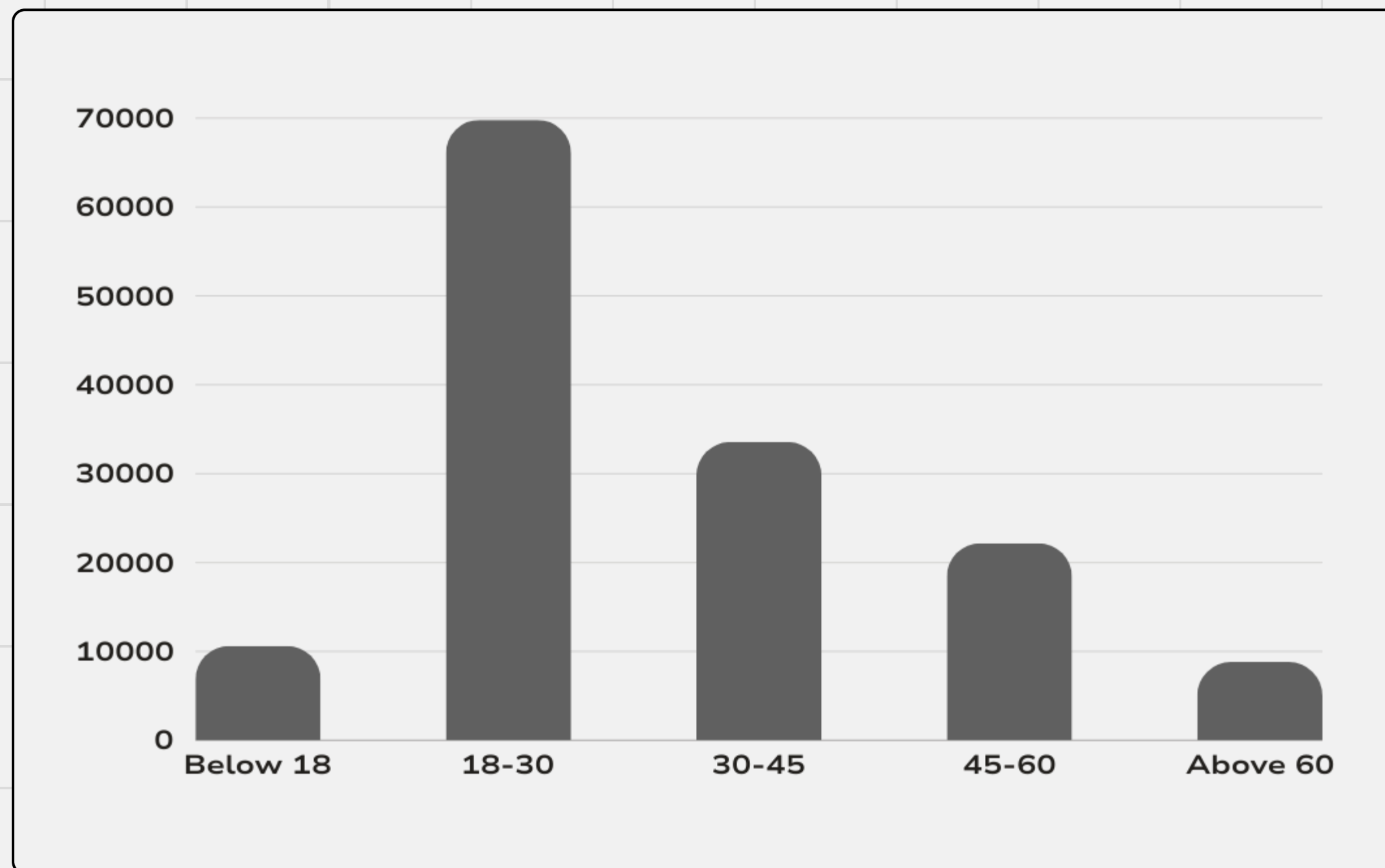
TEAM-T212

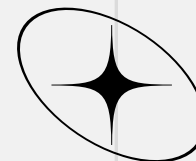




PROBLEM STATEMENT

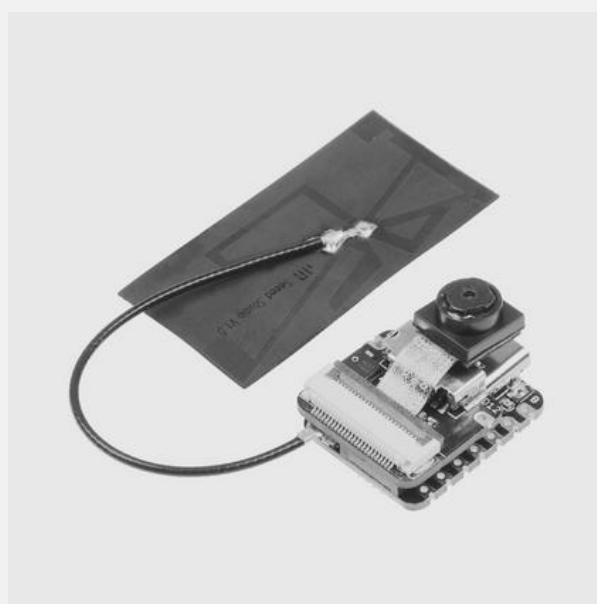
Majority of accidents are caused due to the lack of importance given to the use of helmets.

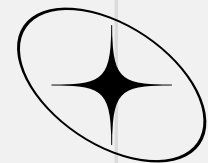




IDEA

Utilising a facial recognition system, it identifies the presence or absence of the helmet.





WORKING



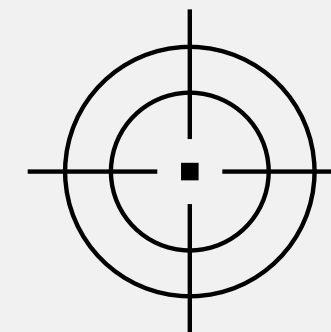
EDGE IMPULSE AND ARDUINO

- Trained AI to recognise the presence or absence of helmet



XIAO ESP32S3

- Code is uploaded
- Camera captures images



PRODUCT

- Keyhole can be opened or closed depending on whether the helmet is detected or not.

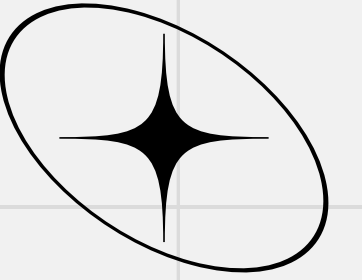
EDGE IMPULSE

- Machine learning software where we can train the system to detect the helmet.

The screenshot displays the Edge Impulse Studio web interface in a browser window. The URL is `studio.edgeimpulse.com/studio/345530/acquisition/training?page=1`. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Impulse design, and EON Tuner. Under Data acquisition, there are options for Create impulse, Image, and Object detection. At the bottom, there is a 'Try Enterprise Free' section with a 'Start free trial' button.
- Top Bar:** Shows 'Dataset', 'Data sources', and 'Labeling queue (0)'.
- Main Content Area:**
 - Dataset Summary:** Displays 'DATA COLLECTED 193 items' and 'TRAIN / TEST SPLIT 79% / 21%'.
 - Dataset Table:** A table with columns 'SAMPLE NAME', 'LABELS', and 'ADDED'. It lists training samples (152) and test samples (41). The sample `helmet.4lrhh8f1` is highlighted.
 - Collect data:** A section with a 'Connect a device' button to start building the dataset.
 - RAW DATA:** A section showing a raw image of a person wearing a helmet, with a red bounding box labeled 'helmet'.
 - Metadata:** A section showing 'No metadata'.

The Windows taskbar at the bottom shows the time as 15:35 on 07-02-2024, with various system icons and a search bar.



ARDUINO IDE

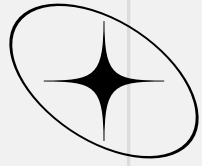


- It is an open source software used for writing compiling and uploading to almost all arduino modules.

XIAO ESP32S3 SENSE



The camera module embedded in the system helps in capturing the images of the helmet.



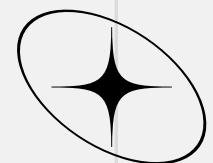
PRODUCT

- Keyhole can be opened or closed depending on whether the helmet is detected or not.



- ROTARY BIKE SWITCH





IR FILTER

can be used to
capture images at
night

INCREASING ACCURACY

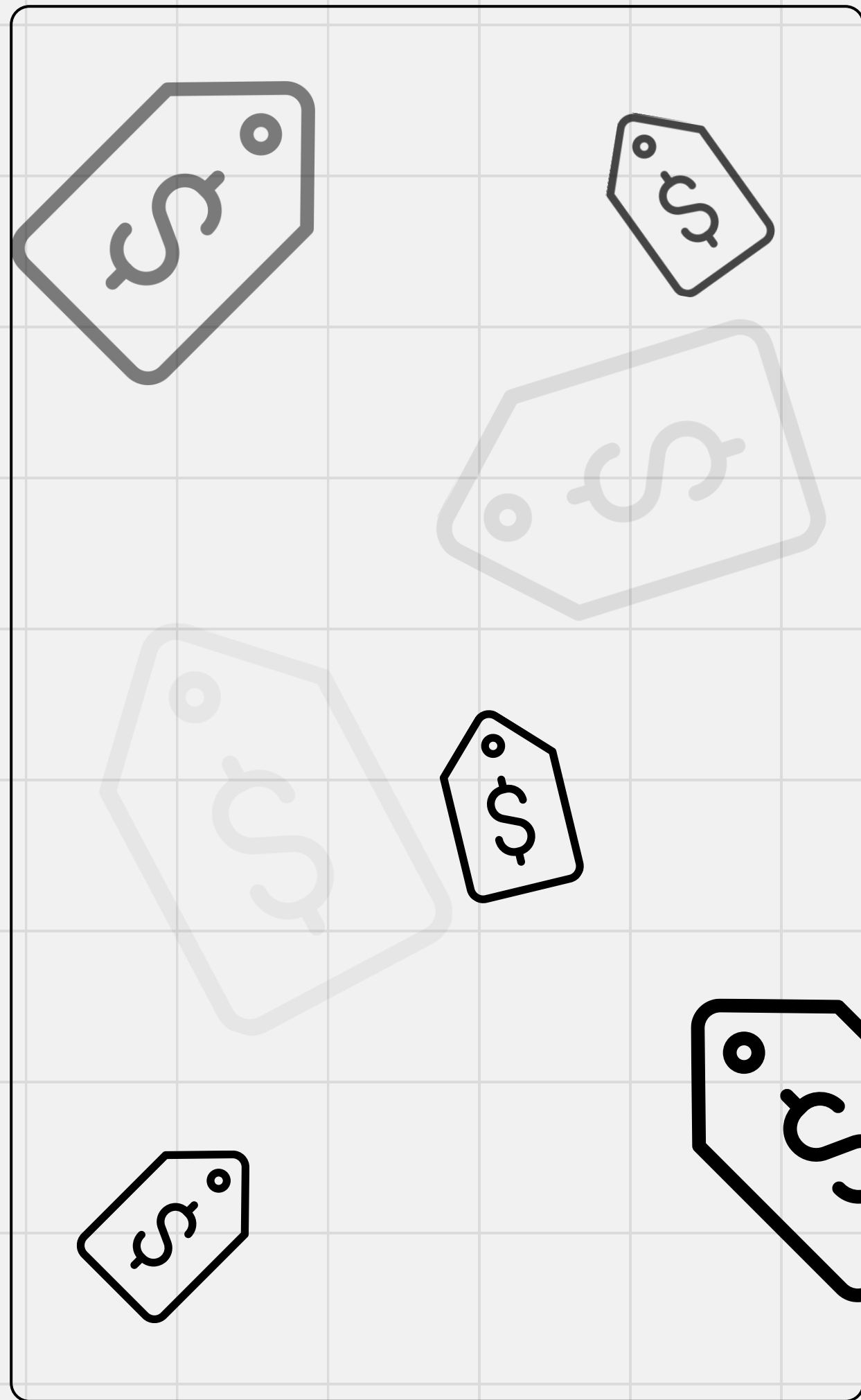
Training ML with
improved data sets for
increased accuracy

DIRECT CONNECTION TO ENGINE

ALARM

As an alternative to
the LED, an alarm can
be set up

MODIFICATIONS



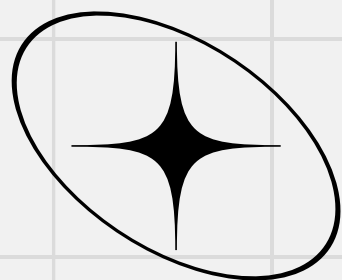
COST OF MAKING

- XIAO ESP32S3 SENSE BOARD:1500/-
 - OTHERS: 100/-
 - TOTAL COST: 1600/-
-

SOCIAL IMPACT

- reduce the risk of head injuries
- promotes better road culture





THANK YOU

